

Phase space mixing in the presence of self-gravity: spherically symmetric Vlasov-Poisson evolutions with fixed and dispersed angular momentum

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(Dated: October 4, 2026)

[[Erik] Provisional; to be rewritten when sections VI and VII are final.] This article is devoted to the numerical study of a spherically symmetric, collisionless kinetic gas which is trapped in an external central potential and whose self-gravity is taken into account. In the absence of self-gravity, such a gas relaxes in time to a stationary state through phase space mixing. We analyze how this process is modified by the self-gravity of the gas, considering both the reduced model in which all gas particles have the same magnitude of the angular momentum and the case in which the angular momenta are distributed over an interval. Our study is based on a particle-in-cell code, on a numerical solution of the linearized equations in action-angle variables, and on a spectral criterion which characterizes the existence of oscillating modes. We find that small perturbations of a self-consistent steady state are damped as long as the mass of the gas lies below a threshold, whereas above this threshold they give rise to an oscillation which is not damped, and we show that the threshold and the frequency of the oscillation are accurately predicted by the spectral criterion. Finally, we show that a dispersion in the angular momentum suppresses the oscillations which are present in the reduced model.

PACS numbers: 05.20.Dd, 98.10.+z, 02.60.Cb

I. INTRODUCTION

Collisionless kinetic gases provide the standard description for systems consisting of a large number of particles which interact with each other only through the mean field they generate. Prominent examples are the stars in a galaxy or in a globular cluster, and the dark matter particles in a galactic halo [1]. The state of such a gas at time t is described by its one-particle distribution function (DF) $f(t, x, p)$, a nonnegative function on the one-particle phase space which determines the number of gas particles in a small phase space volume about the point (x, p) , and whose evolution is governed by the Vlasov equation.¹ In the Newtonian gravitational case, the Vlasov equation is coupled to Poisson's equation for the gravitational potential generated by the gas, which yields the Vlasov-Poisson system, see Ref. [2] for a review of its mathematical properties. Its general relativistic counterpart is the Vlasov-Einstein system [3].

It is instructive to compare the dynamics of a self-gravitating collisionless gas with the one of a pressureless fluid (dust), which is obtained in the limit in which the velocity dispersion of the gas vanishes. The gravitational collapse of a spherical dust cloud leads to the formation of a singularity in finite time at which the density diverges, both in the Newtonian theory and in general relativity [4]; in the latter case, part of this singularity may even be visible to distant observers [5, 6]. In contrast to this, the velocity dispersion of a kinetic gas prevents the particles from focusing. In fact, the Vlasov-Poisson system admits global classical solutions for general initial data [7–9], while for the spherically symmetric Vlasov-Einstein system global solutions exist for small initial data [10] and sufficiently concentrated data collapse to a black hole [11], the threshold between these two outcomes having been explored numerically in Refs. [12–15]. Therefore, for a Newtonian collisionless gas the relevant question is not whether or not a singularity forms, but rather how the gas behaves for large times and, in particular, whether or not it relaxes to a stationary state.

A fundamental mechanism leading to such a relaxation is phase space mixing [16, 17]. To explain it, consider first a gas whose self-gravity is neglected and which is trapped in the potential well of an external potential. If the one-particle Hamiltonian is integrable, each gas particle is confined to an invariant torus, and in terms of action-angle variables (Q, J) the solution of the Vlasov equation is $f(t, Q, J) = F(Q - \Omega(J)t, J)$, where F describes the initial DF and $\Omega(J)$ denotes the frequencies associated with the torus labeled by J . Since these frequencies vary from one torus to the other, the DF is stretched along the angle directions and develops structures on finer and finer scales as time increases. As a consequence, f does not converge pointwise as $t \rightarrow \infty$. However, the macroscopic observables which are obtained by integrating f against a test function do converge, and their limit is the one associated with

¹ The Vlasov equation is also often called the collisionless Boltzmann equation in the literature.

the average of the initial DF over the angle variables, which depends only on integrals of motion. A mathematically precise formulation of this statement and its proof for rotationally symmetric potentials in $d = 1, 2, 3$ space dimensions satisfying a suitable non-degeneracy condition were given by Rioseco and Sarbach [18] (see also Refs. [19, 20]). The results in Ref. [18] include the case of a spherically symmetric gas configuration in which all the gas particles have the same value of the total angular momentum, which behaves like an effective one-dimensional system, and they explain the relaxation to a virialized state which had been observed in the numerical simulations of Ref. [21]. For the analogous phenomenon in the exterior of a black hole we refer the reader to Refs. [22, 23], and for estimates on the rate at which the observables converge to Refs. [24–26].

The generalization to the self-gravitating case, which was left open in Ref. [18], is considerably more involved, since the potential now depends on the DF, such that a perturbation of the latter induces a perturbation of the potential which, in turn, acts back on the gas. In plasma physics, where the particles interact through their mean electric field, the corresponding collective effect has been known since the work of Landau [27]: for a gas of charged particles close to a spatially homogeneous equilibrium the electric field decays in time while the DF undergoes mixing. This phenomenon, known as Landau damping, has been established rigorously at the nonlinear level by Mouhot and Villani [28]. The gravitational case differs from the electromagnetic one in two important aspects. First, the interaction is attractive, such that the collective response of the gas tends to reinforce a perturbation instead of screening it. Second, a self-gravitating gas configuration of finite mass does not possess spatially homogeneous equilibria. Instead, its steady states are spatially inhomogeneous, the gas particles are trapped in the potential well, and the frequencies of their radial motion lie in an interval which, in the situations considered in this article, is bounded and bounded away from zero. For this reason, a collective oscillation whose frequency lies below this interval is not in resonance with any gas particle and hence cannot be damped by mixing.

The question of whether small perturbations of a steady state of the gravitational Vlasov-Poisson system are damped or give rise to persistent oscillations goes back to Lynden-Bell [16] and has received considerable attention in recent years. Hadžić, Rein and Straub [29] (see also Refs. [30, 31]) showed that the essential spectrum of the operator governing the linearized dynamics about a large class of spherically symmetric steady states is determined by the radial periods of the gas particles. Using a version of the Birman-Schwinger principle which was first applied to this problem by Mathur [32], they derived a criterion for the existence of an eigenvalue in the gap below this spectrum, which corresponds to an oscillating mode that is not damped. This criterion is formulated in terms of a family of bounded operators acting on functions of the radius. Related criteria for the existence of oscillating modes and estimates for their number, which also allow for an attractive external potential, have been given by Moreno, Rioseco and Van Den Bosch [33] for steady states whose DF depends only on the energy. For the reduced system in which all the gas particles have the same magnitude of the angular momentum and move around a central point mass, Hadžić, Rein, Schrecker and Straub [34] proved the following dichotomy for steady states of sufficiently small mass whose DF is proportional to $(E_0 - E)^k$ for energies E below the cutoff energy E_0 : the solutions of the linearized system are damped if $k > 1$, whereas an oscillating mode exists if $1/2 < k \leq 1$. Quantitative decay rates for the linearized system about steady states with a central point mass and a distribution of angular momenta have been obtained in Ref. [35], and for the nonlinear system with an external Kepler potential, phase mixing of spherically symmetric data over long time scales has been established in Ref. [36]. On the numerical side, Ramming and Rein [37] and Straub [38] have performed particle-in-cell simulations of the spherically symmetric Vlasov-Poisson system which show that, depending on the steady state, its perturbations either oscillate without damping or decay. Finally, we mention that weakly damped non-radial modes of spherical stellar systems have been computed in Refs. [39, 40] by means of the matrix method.

In this article, we study numerically the transition between mixing and persistent oscillations for a spherically symmetric collisionless gas which is trapped in an external central potential and whose self-gravity is taken into account. We consider two situations. The first one is the reduced model in which all the gas particles have the same magnitude $L_0 > 0$ of the angular momentum, which has a two-dimensional phase space and is the setting considered in Refs. [18, 34]. In the second situation the angular momenta of the gas particles are distributed over an interval. For the external potential we choose the isochrone potential, for which the action-angle variables are known explicitly, and the potential generated by a point mass. Our analysis is based on three independent tools: (i) a particle-in-cell (PIC) code which solves the nonlinear Vlasov-Poisson system, (ii) a numerical solution of the equations linearized about a self-consistent steady state, formulated in the action-angle variables of the latter, and (iii) the numerical computation of the largest eigenvalue $\lambda(\omega)$ of the operator of Mathur, which measures the gain of the self-consistent loop in which a perturbation of the potential with frequency ω induces a perturbation of the density which, in turn, generates a potential. An oscillating mode with frequency below the frequency band of the gas particles exists if and only if the value λ_{edge} of this gain at the lower edge of the band is larger than one.

The main results of this article can be summarized as follows. First, as long as the mass of the gas is sufficiently small compared to the mass which generates the external potential, a small perturbation of a self-consistent steady state is damped, and the frequency and the damping rate of the collective response obtained from the PIC simulations

agree with those obtained from the linearized equations to within one percent. Second, for the reduced model there is a threshold mass above which a discrete mode emerges below the frequency band, such that the perturbation oscillates without being damped. We show that this threshold is characterized by the condition $\lambda_{edge} = 1$, and that the frequency of the mode predicted by the equation $\lambda(\omega) = 1$ agrees with the one measured in the simulations. Third, for steady states of small mass with $k \leq 1$, the oscillating mode whose existence is guaranteed by the results in Ref. [34] lies so close to the edge of the band that it cannot be distinguished from the decaying part of the response over the time span of our simulations; instead, we observe an algebraic decay proportional to $t^{-(k+1)}$ for all the values of k we have considered. Fourth, a dispersion in the angular momentum removes the divergence of $\lambda(\omega)$ at the edge of the band which is responsible for the oscillating modes with $k \leq 1$ in the reduced model, such that these modes disappear when the mass of the gas is small enough, and it suppresses the coherent response which is present when all the gas particles have the same angular momentum. **[(Erik) The last statement relies on the operator with dispersion and on the free solution; the PIC runs with dispersion of section VII are still to be done.]**

Particle methods have a long tradition in the numerical study of spherically symmetric collisionless systems, both in the Newtonian case [37, 38] and in the relativistic one, where they have been used to analyze the collapse of star clusters and the critical behavior at the threshold of black hole formation [12–14, 41, 42]; see Refs. [43–45] for introductions to the PIC method, Ref. [46] for a higher-order implementation in the context of plasma physics, and Ref. [47] for a convergence proof of a PIC scheme for the spherically symmetric Vlasov-Einstein system. Measuring the response of the gas to a small perturbation with such a method is delicate, because the signal has to be separated from the noise which is due to the finite number of computational particles. In this article we address this problem by placing the particles on a regular lattice in the action-angle variables of the self-consistent steady state, by encoding the perturbation in the weights of the particles instead of their positions (as in Refs. [48, 49]), and by subtracting a reference run without perturbation. We also show that the observables need to be computed with the action-angle variables belonging to the total potential, since those of the external potential alone leave a time-independent residue which could be mistaken for an undamped mode.

The remainder of this article is organized as follows. In section II we formulate the Vlasov-Poisson system with an external potential in spherical symmetry, for the reduced model with fixed angular momentum as well as for a distribution of angular momenta, and we introduce the action-angle variables and the observables used throughout this article. In section III we review the mixing phenomenon in the absence of self-gravity, derive the linearized equations about a steady state and introduce the gain $\lambda(\omega)$ of the self-consistent loop. Our numerical methods are described in section IV, and their validation is presented in section V. The results for the reduced model are discussed in section VI, and those for a distribution of angular momenta in section VII. Conclusions and an outlook to future work are given in section VIII. Technical details regarding the numerical computation of the action-angle variables are included in Appendix A.

II. THE VLASOV-POISSON SYSTEM IN SPHERICAL SYMMETRY

In this section we formulate the Vlasov-Poisson system with an external potential and reduce it to spherical symmetry, first for a gas whose particles have arbitrary angular momenta and then for the reduced model in which all of them have the same magnitude of the angular momentum. Next, we introduce the dimensionless variables used in our numerical computations, the action-angle variables associated with a time-independent potential and, finally, the observables on which our analysis of the mixing phenomenon is based.

We consider a collisionless gas consisting of identical particles which are subject to a time-independent external potential Φ_{ext} and to the gravitational potential Φ_{gas} generated by the gas itself. Since all the gas particles have the same mass, it is convenient to normalize the DF such that $f(t, x, p)d^3x d^3p$ is the mass of the gas contained in the volume element $d^3x d^3p$ of phase space, where from now on p denotes the momentum per unit mass.² With these conventions, the Vlasov-Poisson system consists of the Vlasov equation

$$\frac{\partial f}{\partial t} + p \cdot \nabla_x f - (\nabla_x \Phi) \cdot \nabla_p f = 0, \quad \Phi := \Phi_{ext} + \Phi_{gas}, \quad (1)$$

and the Poisson equation

$$\Delta \Phi_{gas} = 4\pi G \rho, \quad \rho(t, x) := \int_{\mathbb{R}^3} f(t, x, p) d^3p, \quad (2)$$

² This is equivalent to working in units for which the mass of each gas particle is one, as in Ref. [18].

where G denotes Newton's constant and ρ is the mass density of the gas. To obtain a unique solution of Eq. (2) we require that $\Phi_{gas}(t, x) \rightarrow 0$ for $|x| \rightarrow \infty$.

For the external potential we choose Hénon's isochrone potential [1, 50, 51]

$$\Phi_{ext}(x) = -\frac{GM}{b + \sqrt{b^2 + r^2}}, \quad r := |x|, \quad (3)$$

which is generated by a spherical matter distribution of total mass M . The parameter $b \geq 0$ is a length scale which describes the transition from a harmonic potential (when $r \ll b$) to the potential of a point particle of mass M (when $r \gg b$). For $b = 0$, Eq. (3) reduces to the Kepler potential $\Phi_{ext} = -GM/r$, which is the external potential considered in Ref. [34]. The reason for this choice is that in this potential the period of the radial motion depends only on the energy of the particle [50, 51], and that the corresponding action-angle variables are known in closed form [1]. When the self-gravity of the gas is neglected, this provides exact solutions against which our numerical methods can be validated.

A. Reduction to spherical symmetry

A spherically symmetric gas configuration is described by a DF which is invariant with respect to simultaneous rotations of x and p . Such a DF depends on (x, p) only through the radius r , the radial component $p_r := x \cdot p/r$ of the momentum and the magnitude $L := |x \wedge p|$ of the angular momentum, that is,

$$f(t, x, p) = \mathcal{F}(t, r, p_r, L). \quad (4)$$

In this case, the density ρ and the potential Φ_{gas} depend only on t and r , and the Vlasov equation (1) reduces to

$$\frac{\partial \mathcal{F}}{\partial t} + p_r \frac{\partial \mathcal{F}}{\partial r} - \frac{\partial V_L}{\partial r} \frac{\partial \mathcal{F}}{\partial p_r} = 0, \quad V_L(t, r) := \Phi(t, r) + \frac{L^2}{2r^2}, \quad (5)$$

with V_L the effective potential for the radial motion. Since no derivative with respect to L appears in Eq. (5), the gas particles with a given value of L behave like a one-dimensional gas in the potential V_L . However, particles with different values of L interact with each other through the potential Φ_{gas} , which is determined by the spherically reduced Poisson equation

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi_{gas}}{\partial r} \right) = 4\pi G \rho, \quad \rho(t, r) = \frac{2\pi}{r^2} \int_0^\infty \int_{-\infty}^\infty \mathcal{F}(t, r, p_r, L) dp_r L dL, \quad (6)$$

see for instance Ref. [21]. The solution of Eq. (6) which is regular at the center and vanishes at infinity is

$$\Phi_{gas}(t, r) = -4\pi G \int_0^\infty \frac{\rho(t, s)}{\max\{r, s\}} s^2 ds, \quad (7)$$

and its gradient is $\partial \Phi_{gas} / \partial r = Gm(t, r)/r^2$, where $m(t, r) := 4\pi \int_0^r \rho(t, s) s^2 ds$ is the mass of the gas contained in the sphere of radius r at time t .

For the following, it is useful to note that when a spherically symmetric function is integrated over phase space, the volume element can be replaced with

$$d^3x d^3p = 8\pi^2 dr dp_r L dL. \quad (8)$$

In particular, the total mass and the total energy of the gas are given by

$$M_{gas} = 8\pi^2 \int_0^\infty \int_{-\infty}^\infty \int_0^\infty \mathcal{F}(t, r, p_r, L) dr dp_r L dL, \quad (9)$$

$$\mathcal{E}_{gas} = 8\pi^2 \int_0^\infty \int_{-\infty}^\infty \int_0^\infty \mathcal{F}(t, r, p_r, L) \left[\frac{p_r^2}{2} + \frac{L^2}{2r^2} + \Phi_{ext}(r) + \frac{1}{2} \Phi_{gas}(t, r) \right] dr dp_r L dL, \quad (10)$$

both of which are conserved in time. The conservation of $\mathcal{E}_{gas}$ provides a test for the accuracy of our numerical evolutions, see section V.

B. The reduced model with fixed angular momentum

A further simplification, which has been considered in Refs. [18, 21, 34], is obtained by assuming that all the gas particles have the same magnitude $L_0 > 0$ of the angular momentum.³ In this case, the DF is of the form

$$\mathcal{F}(t, r, p_r, L) = \mathcal{F}_0(t, r, p_r) \delta(L - L_0), \quad (11)$$

and Eqs. (5,6) yield

$$\frac{\partial \mathcal{F}_0}{\partial t} + p_r \frac{\partial \mathcal{F}_0}{\partial r} - \frac{\partial V_{L_0}}{\partial r} \frac{\partial \mathcal{F}_0}{\partial p_r} = 0, \quad \rho(t, r) = \frac{2\pi L_0}{r^2} \int_{-\infty}^{\infty} \mathcal{F}_0(t, r, p_r) dp_r, \quad (12)$$

where the potential Φ_{gas} is still given by Eq. (7). Likewise, Eqs. (9,10) hold with $\mathcal{F}$ replaced by $\mathcal{F}_0$, L by L_0 , and the integral over L by the factor L_0 . Equations (12,7) constitute a Vlasov-Poisson system on a two-dimensional phase space with coordinates (r, p_r) . It describes a gas of concentric spherical shells which move in the effective potential V_{L_0} . The centrifugal term in this potential prevents the shells from reaching the center, and the gas configurations considered in this article are confined to a region of the form $0 < r_{in} \leq r \leq r_{out}$. In spite of its simplicity, the reduced model retains the two ingredients which are relevant for this article, namely the dependency of the frequency of the radial motion on the orbit, which is responsible for the mixing, and the self-gravity of the gas.

In the second situation considered in this article, $\mathcal{F}$ is a regular function of L whose support is contained in an interval $[L_1, L_2]$ with $0 < L_1 < L_2$. In this case, the phase space is three-dimensional, and the orbits with different values of L have different frequencies.

C. Dimensionless variables

In our numerical computations we use dimensionless variables, which are obtained by measuring masses in units of M , lengths in units of b and times in units of the dynamical time

$$t_{dyn} := \sqrt{\frac{b^3}{GM}}. \quad (13)$$

Accordingly, p_r is measured in units of $\sqrt{GM/b}$, L in units of $\sqrt{GMb}$, the potentials and the energy per unit mass in units of GM/b , and $\mathcal{F}$ in units of $M/(Gmb)^{3/2}$. In terms of these variables, all the equations of this section retain their form with $G = M = b = 1$; in particular, the isochrone potential is $\Phi_{ext}(r) = -1/(1 + \sqrt{1 + r^2})$. The resulting system contains no free parameters, and the strength of the self-gravity is controlled by the ratio M_{gas}/M between the mass of the gas and the mass which generates the external potential. In the limit $M_{gas}/M \rightarrow 0$ the potential Φ_{gas} can be neglected, and one recovers the model analyzed in Ref. [18].

In the Kepler case $b = 0$ the external potential does not provide a length scale. However, in this case the system (5,6) is invariant with respect to the rescaling

$$(t, r, p_r, L, \mathcal{F}, \Phi) \mapsto (\alpha^{3/2}t, \alpha r, \alpha^{-1/2}p_r, \alpha^{1/2}L, \alpha^{-3/2}\mathcal{F}, \alpha^{-1}\Phi), \quad \alpha > 0, \quad (14)$$

which leaves M_{gas} unchanged. Therefore, b can be replaced with an arbitrary length in the definitions above, such that $\Phi_{ext}(r) = -1/r$, and in the reduced model this freedom can be used to fix the value of L_0 .

To give an idea of the physical time scale, we note that $M = 10^5 M_\odot$ and $b = 1$ pc, which are of the order of magnitude of the mass and of the core radius of a globular cluster [1], yield $t_{dyn} \approx 4.7 \times 10^4$ yr. As will be seen below, $2\pi t_{dyn}$ is a lower bound for the periods of the radial motion in the isochrone potential.

D. Steady states and action-angle variables

Suppose that the total potential Φ is independent of time. Then, the energy per unit mass

$$E := \frac{p_r^2}{2} + V_L(r) \quad (15)$$

³ Note that in Ref. [34] the symbol L refers to the square of the magnitude of the angular momentum.

is conserved along the trajectories of the gas particles, and it follows that any DF of the form

$$\mathcal{F}(r, p_r, L) = \psi(E, L), \quad (16)$$

with a nonnegative function ψ , is a time-independent solution of the Vlasov equation (5). In order for Eq. (16) to provide a solution of the Vlasov-Poisson system, the potential Φ_{gas} , which enters the definition of E through V_L , has to coincide with the one generated by the density of the DF (16) through Eqs. (6,7). This is a nonlinear problem for Φ_{gas} , whose numerical solution is described in section IV. In the following, a solution $(\mathcal{F}_{eq}, \Phi_{eq})$ of this problem, with Φ_{eq} the total potential and ρ_{eq} the corresponding density, will be referred to as a self-consistent steady state.

Next, we introduce action-angle variables for the radial motion in a time-independent potential Φ , following Refs. [1, 18]. In our applications, Φ is either the external potential (3) or the total potential Φ_{eq} of a self-consistent steady state. In both cases $\Phi'(r) = Gm_{tot}(r)/r^2$, where $m_{tot}(r)$ denotes the total mass contained in the sphere of radius r , including the one which generates the external potential. Since the function $r \mapsto rm_{tot}(r)$ increases strictly monotonically from 0 to ∞ , for each $L > 0$ the effective potential V_L has a unique critical point $r = r_{min}(L)$. It is determined by the equation $r^3\Phi'(r) = L^2$, it is a non-degenerate minimum of V_L , and it corresponds to a circular orbit with energy $E_{min}(L) := V_L(r_{min}(L))$. Since $V_L(r) \rightarrow \infty$ for $r \rightarrow 0$ and $V_L(r) \rightarrow 0$ for $r \rightarrow \infty$, the orbits with energies in the range $E_{min}(L) < E < 0$ are bound, and they oscillate between the turning points $r_1(E, L) < r_2(E, L)$, which are the two solutions of the equation $V_L(r) = E$.

As in Ref. [18] we define the area function

$$A(E, L) := \oint p_r dr = 2 \int_{r_1(E, L)}^{r_2(E, L)} \sqrt{2(E - V_L(r))} dr, \quad (17)$$

which represents the area enclosed by the orbit in the (r, p_r) -plane, and whose partial derivative with respect to E yields the period of the radial motion,

$$T(E, L) := \frac{\partial A}{\partial E}(E, L) = 2 \int_{r_1(E, L)}^{r_2(E, L)} \frac{dr}{\sqrt{2(E - V_L(r))}}. \quad (18)$$

The action and angle variables associated with the radial motion are

$$J := \frac{A(E, L)}{2\pi}, \quad Q := \frac{2\pi}{T(E, L)} \int_{\gamma_r} \frac{dr}{p_r}, \quad (19)$$

where γ_r denotes the segment of the orbit in the (r, p_r) -plane which connects, in the clockwise direction, the inner turning point $(r_1(E, L), 0)$ with the given point (r, p_r) . Hence, Q/Ω with $\Omega := 2\pi/T(E, L)$ is the time which has elapsed since the last passage of the particle through its pericenter; in particular, $Q = 0$ at the pericenter, $Q = \pi$ at the apocenter, and $0 < Q < \pi$ when $p_r > 0$. For each fixed value of L , the map $(r, p_r) \mapsto (Q, J)$ is symplectic, that is $dr \wedge dp_r = dQ \wedge dJ$, and it provides smooth coordinates on the set of bound orbits which are not circular. Since $T > 0$, the first relation in Eq. (19) can be inverted for each L , which yields the energy as a function $E(J, L)$ of the action variable and the angular momentum, and the frequency of the radial motion is

$$\Omega(J, L) = \frac{\partial E}{\partial J}(J, L) = \frac{2\pi}{T(E, L)}. \quad (20)$$

[(Erik) We denote the orbital frequency by Ω (it is called ω in Ref. [18]) and reserve ω for the frequency of the collective oscillation.] In terms of the variables (Q, J, L) , the volume element (8) is $8\pi^2 dQ dJ L dL$, and the Vlasov equation (5) for a time-independent potential assumes the simple form

$$\frac{\partial \mathcal{F}}{\partial t} + \Omega(J, L) \frac{\partial \mathcal{F}}{\partial Q} = 0. \quad (21)$$

In particular, the steady states (16) are precisely those DFs which are independent of the angle variable, and they can be regarded as functions of (J, L) .

For the external potential (3) the area function can be computed explicitly, see Refs. [1, 18]. In terms of the dimensionless variables introduced in the previous subsection, one finds

$$A(E, L) = 2\pi \left[\frac{1}{\sqrt{-2E}} - c(L) \right], \quad E_{min}(L) = -\frac{1}{2c(L)^2}, \quad (22)$$

with $c(L) := (L + \sqrt{L^2 + 4})/2$ for the isochrone potential and $c(L) := L$ for the Kepler potential. This yields

$$E(J, L) = -\frac{1}{2[J + c(L)]^2}, \quad \Omega(J, L) = \frac{1}{[J + c(L)]^3} = (-2E)^{3/2}. \quad (23)$$

Therefore, the frequency depends only on the energy, it decreases monotonically when J or L increase, and for each fixed L its maximum is attained in the limit $J \rightarrow 0$ of circular orbits. For the isochrone potential, $c(L) > 1$ implies that $\Omega < 1$, such that the periods of the radial motion are larger than 2π . An explicit expression for the angle variable Q in terms of (r, p_r, L) can be found in section 3.5.2 of Ref. [1]; its inversion requires the solution of an equation of Kepler's type. In contrast to this, for a self-consistent steady state the potential Φ_{eq} is only known numerically, and in this case we compute the action-angle variables by numerical quadrature of Eqs. (17,18,19), see Appendix A.

E. Observables

As in Ref. [18], we describe the macroscopic state of the gas through observables of the form

$$N_\varphi(t) := \int f(t, x, p) \varphi(x, p) d^3x d^3p, \quad (24)$$

where φ is a given test function on phase space. In this article we consider two classes of such observables.

The first one refers to the action-angle variables (Q, J) belonging to a time-independent reference potential. Unless otherwise stated, this is the total potential Φ_{eq} of the self-consistent steady state whose perturbations are analyzed, which reduces to Φ_{ext} when the self-gravity of the gas is neglected. For a spherically symmetric configuration and a test function of the form $\varphi = a(Q)B(J, L)$, with real-valued functions a and B , Eqs. (8,24) and the expansion of a in a Fourier series yield

$$N_\varphi(t) = \sum_{n \in \mathbb{Z}} \hat{a}_n^* h_n(t), \quad \hat{a}_n := \frac{1}{2\pi} \int_0^{2\pi} a(Q) e^{-inQ} dQ, \quad (25)$$

with the functions

$$h_n(t) := 8\pi^2 \int_0^\infty \int_0^\infty \int_0^{2\pi} \mathcal{F}(t, Q, J, L) B(J, L) e^{-inQ} dQ dJ L dL, \quad n \in \mathbb{Z}. \quad (26)$$

In the reduced model, B depends only on J and Eq. (26) holds with $\mathcal{F}$ replaced by $\mathcal{F}_0$ and the integral over L by the factor L_0 . Since $\mathcal{F}$ is real, $h_{-n} = h_n^*$, such that it is sufficient to consider $n \geq 0$. For a steady state h_n vanishes for all $n \neq 0$, whereas h_0 is equal to the mass of the gas weighted with the function B . Therefore, the functions $h_n(t)$ with $n \neq 0$ measure the departure of the gas configuration from a steady state, and their decay in time is the signature of the relaxation process we are interested in. We stress that this property relies on the use of the action-angle variables which belong to the total potential Φ_{eq} . If those of the external potential are used instead, a self-consistent steady state gives rise to nonvanishing, time-independent values of h_n with $n \neq 0$, as we show in section V.

The second class of observables consists of the perturbation of the gravitational potential,

$$\delta\Phi(t, r) := \Phi(t, r) - \Phi_{eq}(r) = -4\pi G \int_0^\infty \frac{\rho(t, s) - \rho_{eq}(s)}{\max\{r, s\}} s^2 ds, \quad (27)$$

where we have used Eq. (7). For each fixed r this is the difference between the values of the observable (24) with test function $\varphi(x, p) = -G/\max\{r, |x|\}$ which belong to the DF of the gas and to the one of the steady state. In contrast to the functions $h_n(t)$, the quantity $\delta\Phi$ does not refer to any particular coordinates on phase space. The results in Refs. [34, 35] refer to the decay of the force field $-\partial\delta\Phi/\partial r$ which is generated by it, or of the time derivative of this field. To measure its magnitude we use the root mean square

$$\|\delta\Phi(t)\| := \left(\frac{1}{r_b - r_a} \int_{r_a}^{r_b} \delta\Phi(t, r)^2 dr \right)^{1/2} \quad (28)$$

over an interval $[r_a, r_b]$ which covers the region occupied by the gas.

III. MIXING AND LINEAR RESPONSE

In this section we first review the mixing phenomenon in the absence of self-gravity and discuss the rate at which the observables introduced in the previous section decay. Next, we linearize the Vlasov-Poisson system about a self-consistent steady state and formulate, in terms of the action-angle variables of the latter, the criterion for the existence of oscillating modes. This leads to the gain $\lambda(\omega)$ of the self-consistent loop mentioned in the introduction, whose behavior at the edge of the frequency band is discussed at the end of the section. The derivations in this section are formal; for rigorous statements we refer the reader to Refs. [18, 20] regarding the mixing property and to Refs. [29, 30, 33, 34] regarding the linearized system.

A. Mixing in the absence of self-gravity

When the self-gravity of the gas is neglected, $\Phi = \Phi_{ext}$ is independent of time, and the solution of Eq. (21) is

$$\mathcal{F}(t, Q, J, L) = F(Q - \Omega(J, L)t, J, L), \quad (29)$$

where the function F , which is 2π -periodic in its first argument, describes the initial DF. Expanding F in a Fourier series with respect to the angle variable,

$$F(Q, J, L) = \sum_{n \in \mathbb{Z}} \hat{F}_n(J, L) e^{inQ}, \quad \hat{F}_n(J, L) := \frac{1}{2\pi} \int_0^{2\pi} F(Q, J, L) e^{-inQ} dQ, \quad (30)$$

and introducing Eq. (29) into Eq. (26) one obtains

$$h_n(t) = 16\pi^3 \int_0^\infty \int_0^\infty \hat{F}_n(J, L) B(J, L) e^{-in\Omega(J, L)t} dJ L dL. \quad (31)$$

Hence, h_0 is constant in time, whereas for $n \neq 0$ the function $h_n(t)$ is a superposition of oscillations whose frequencies $n\Omega(J, L)$ vary from one orbit to the other. Provided that the gradient of Ω with respect to (J, L) is different from zero for almost all (J, L) (in the reduced model, provided that $\partial\Omega/\partial J \neq 0$ for almost all J), it follows from the Riemann-Lebesgue lemma and its generalizations (cf. Refs. [18, 20]) that $h_n(t) \rightarrow 0$ for $t \rightarrow \infty$ and all $n \neq 0$. As a consequence, the observables (25) converge to the ones which belong to the angle-averaged DF $\hat{F}_0(J, L)$, which is a steady state according to the remark below Eq. (21). This is the phase space mixing property. For the isochrone and the Kepler potentials, Eq. (23) gives $\partial\Omega/\partial J = -3[J + c(L)]^{-4} < 0$, such that the non-degeneracy condition is satisfied.

The rate at which $h_n(t)$ decays is not universal. It depends on the regularity of the function $\hat{F}_n B$ and on its behavior at the boundary of its support. If this function is smooth and vanishes together with all its derivatives at the boundary, repeated integration by parts in Eq. (31) shows that $h_n(t)$ decays faster than any inverse power of t . A different situation, which is the relevant one for the steady states considered in this article, arises in the reduced model when the DF has its support in an interval $0 \leq J \leq J_{max}$ and vanishes like a power at the outer edge, such that

$$\hat{F}_n(J) B(J) = (J_{max} - J)^k g(J) \quad (32)$$

with an exponent $k > 0$ and a smooth function g satisfying $g(J_{max}) \neq 0$. Suppose that $\Omega'(J) < 0$ for all $0 < J \leq J_{max}$, as is the case in all the examples of this article, and that g vanishes sufficiently fast for $J \rightarrow 0$. Then, the behavior of the integral in Eq. (31) for large times is determined by the endpoint $J = J_{max}$ (see, for instance, chapter 6 in Ref. [52]), and for $n \geq 1$ one finds

$$h_n(t) \simeq 16\pi^3 L_0 \Gamma(k+1) g(J_{max}) \frac{e^{-in\Omega_{min}t}}{[in|\Omega'(J_{max})|t]^{k+1}}, \quad t \rightarrow \infty, \quad (33)$$

where $\Omega_{min} := \Omega(J_{max})$ is the frequency of the outermost orbit and Γ denotes the Gamma function. Therefore, an edge at which the DF vanishes with the exponent k gives rise to an algebraic tail which is proportional to $t^{-(k+1)}$ and which oscillates with the frequency $n\Omega_{min}$: the less regular the DF is at the edge, the slower is the decay. Algebraic decay laws in related settings have been obtained in Refs. [26, 49].

When the angular momenta of the gas particles are distributed, the integration over L in Eq. (31) provides an additional source of decay, since Ω depends on L as well. For the isochrone and the Kepler potentials, Eq. (23) yields $\partial\Omega/\partial L = c'(L) \partial\Omega/\partial J$ with $1/2 < c'(L) \leq 1$, such that a dispersion in L is almost as efficient for the mixing as a dispersion of the same magnitude in J . This effect is analyzed in section VII.

B. Linearization about a self-consistent steady state

Let $(\mathcal{F}_{eq}, \Phi_{eq})$ be a self-consistent steady state, and let (Q, J) be the action-angle variables which belong to the potential Φ_{eq} , such that $\mathcal{F}_{eq}$ is a function of (J, L) only. We write

$$\mathcal{F} = \mathcal{F}_{eq} + \delta\mathcal{F}, \quad \Phi = \Phi_{eq} + \delta\Phi, \quad (34)$$

with $\delta\Phi$ the perturbation of the potential defined in Eq. (27). The motion of the gas particles is determined by the Hamiltonian $H = E(J, L) + \delta\Phi(t, r(Q, J, L))$, where $r(Q, J, L)$ denotes the radius as a function of the action-angle variables. Since for each fixed L the variables (Q, J) are canonical, the Vlasov equation (5) is equivalent to

$$\frac{\partial\delta\mathcal{F}}{\partial t} + \Omega \frac{\partial\delta\mathcal{F}}{\partial Q} - \frac{\partial\mathcal{F}_{eq}}{\partial J} \frac{\partial\delta\Phi}{\partial Q} = \frac{\partial\delta\Phi}{\partial Q} \frac{\partial\delta\mathcal{F}}{\partial J} - \frac{\partial\delta\Phi}{\partial J} \frac{\partial\delta\mathcal{F}}{\partial Q}, \quad (35)$$

where no approximation has been made so far. The right-hand side is quadratic in the perturbation, and neglecting it yields the linearized equation

$$\frac{\partial\delta\mathcal{F}}{\partial t} + \Omega(J, L) \frac{\partial\delta\mathcal{F}}{\partial Q} - \frac{\partial\mathcal{F}_{eq}}{\partial J}(J, L) \frac{\partial\delta\Phi}{\partial Q} = 0. \quad (36)$$

The potential is determined by Eqs. (6,7,8), which give

$$\delta\Phi(t, r) = -8\pi^2 G \int_0^\infty \int_0^\infty \int_0^{2\pi} \frac{\delta\mathcal{F}(t, Q', J', L')}{\max\{r, r(Q', J', L')\}} dQ' dJ' dL'. \quad (37)$$

The second term on the left-hand side of Eq. (36) describes the transport along the orbits of the steady state discussed in the previous subsection. The third term describes the reaction of the steady state to the perturbation of the potential: the latter changes the action variables of the gas particles at the rate $\dot{J} = -\partial\delta\Phi/\partial Q$, and this displaces mass along the gradient of $\mathcal{F}_{eq}$. Through this term, which is absent when self-gravity is neglected, all the orbits are coupled to each other.

Since the support of $\mathcal{F}_{eq}$ is bounded, the frequencies $\Omega(J, L)$ of the orbits which are populated by the steady state fill an interval $[\Omega_{min}, \Omega_{max}]$ with $0 < \Omega_{min} < \Omega_{max}$, which will be referred to as the frequency band in what follows. Without the third term in Eq. (36), the n -th Fourier coefficient of $\delta\mathcal{F}$ with respect to Q oscillates with the frequencies $n\Omega(J, L)$, which lie in the interval $n[\Omega_{min}, \Omega_{max}]$. These intervals constitute the essential spectrum of the linearized system, which is not altered by the self-gravity of the perturbation [29, 34].

C. Oscillating modes and the gain of the self-consistent loop

We look for solutions of Eqs. (36,37) of the form

$$\delta\mathcal{F}(t, Q, J, L) = a(Q, J, L)e^{-i\omega t} + c.c., \quad \delta\Phi(t, r) = \phi(r)e^{-i\omega t} + c.c., \quad (38)$$

with a real frequency $\omega > 0$, where *c.c.* denotes the complex conjugate of the preceding term. Such a solution describes a perturbation which oscillates without being damped, and it will be called an oscillating mode in what follows. We expand

$$a(Q, J, L) = \sum_{n \in \mathbb{Z}} a_n(J, L)e^{inQ}, \quad \phi(r(Q, J, L)) = \sum_{n \in \mathbb{Z}} \phi_n(J, L)e^{inQ}. \quad (39)$$

Since the radius is the same at the two points of an orbit with angle variables Q and $2\pi - Q$, the coefficients of the potential satisfy $\phi_{-n} = \phi_n$ and are given by

$$\phi_n(J, L) = \frac{1}{2\pi} \int_0^{2\pi} \phi(r(Q, J, L)) \cos(nQ) dQ. \quad (40)$$

Equation (36) yields

$$[n\Omega(J, L) - \omega] a_n(J, L) = n \frac{\partial\mathcal{F}_{eq}}{\partial J}(J, L) \phi_n(J, L), \quad n \in \mathbb{Z}, \quad (41)$$

which implies that $a_0 = 0$. From now on we assume that ω lies below the frequency band, that is $0 < \omega < \Omega_{min}$.⁴ In this case none of the gas particles is in resonance with the perturbation, in the sense that $n\Omega(J, L) \neq \omega$ for all $n \in \mathbb{Z}$ and all the orbits populated by the steady state, and Eq. (41) can be solved for a_n . By the same symmetry argument as above, the potential generated by the perturbation depends only on the combinations $b_n := a_n + a_{-n}$ with $n \geq 1$, for which Eq. (41) gives

$$b_n(J, L) = R_n(J, L; \omega) \phi_n(J, L), \quad R_n(J, L; \omega) := \frac{2n^2 \Omega(J, L)}{n^2 \Omega(J, L)^2 - \omega^2} \frac{\partial \mathcal{F}_{eq}}{\partial J}(J, L). \quad (42)$$

On the other hand, introducing the ansatz (38) into Eq. (37) and using Eq. (40) one obtains

$$\phi_n(J, L) = \sum_{n'=1}^{\infty} \int_0^{\infty} \int_0^{\infty} M_{nn'}(J, L; J', L') b_{n'}(J', L') dJ' L' dL', \quad (43)$$

with the kernel

$$M_{nn'}(J, L; J', L') := -4\pi G \int_0^{2\pi} \int_0^{2\pi} \frac{\cos(nQ) \cos(n'Q')}{\max\{r(Q, J, L), r(Q', J', L')\}} dQ dQ'. \quad (44)$$

Equations (42,43) can be written in the compact form

$$\phi = \mathcal{K}(\omega) \phi, \quad \mathcal{K}(\omega) := M R(\omega), \quad (45)$$

where $\phi = (\phi_n)_{n \geq 1}$, $R(\omega)$ denotes the multiplication with the functions $R_n(\cdot; \omega)$, and M is the integral operator defined by the right-hand side of Eq. (43). The operator $\mathcal{K}(\omega)$ maps a perturbation of the potential with frequency ω to the potential which is generated by the response of the gas to this perturbation. Therefore, an oscillating mode with frequency ω exists if and only if one is an eigenvalue of $\mathcal{K}(\omega)$. In the reduced model, Eqs. (40–44) hold with $L = L' = L_0$, the function $\mathcal{F}_{eq}$ replaced by $\mathcal{F}_{0,eq}(J)$, and the integral over L' by the factor L_0 .

The operators M and $R(\omega)$ have the following properties. First, with respect to the inner product

$$\langle b, c \rangle := \sum_{n=1}^{\infty} \int_0^{\infty} \int_0^{\infty} b_n(J, L) c_n(J, L) dJ L dL, \quad (46)$$

the operator M is symmetric and negative, since for real-valued b one finds

$$-\langle b, Mb \rangle = \frac{1}{64\pi^4 G} \int_{\mathbb{R}^3} |\nabla \delta \Phi_b|^2 d^3x \geq 0, \quad (47)$$

where $\delta \Phi_b$ denotes the potential which is obtained from Eq. (37) when $\delta \mathcal{F}$ is replaced with $\sum_{n \geq 1} b_n(J, L) \cos(nQ)$. Second, suppose that $\partial \mathcal{F}_{eq} / \partial J < 0$ in the interior of the support of the steady state. Since $\partial \mathcal{F}_{eq} / \partial J = \Omega \partial \psi / \partial E$, this means that the function ψ in Eq. (16) decreases with the energy, which is Antonov's stability condition [1, 34], and it holds for all the steady states considered in this article. Then, $R_n(J, L; \omega) < 0$ for all $0 \leq \omega < \Omega_{min}$, and $\mathcal{K}(\omega) = (-M)|R(\omega)|$ has the same nonvanishing eigenvalues as the symmetric and nonnegative operator

$$S(\omega) := |R(\omega)|^{1/2} (-M) |R(\omega)|^{1/2}. \quad (48)$$

We denote by $\lambda(\omega)$ the largest eigenvalue of $S(\omega)$ and call it the gain of the self-consistent loop. Since $|R_n(J, L; \omega)|$ increases with ω , the function $\lambda(\omega)$ is nondecreasing on the interval $[0, \Omega_{min})$, and we define its value at the edge of the band by

$$\lambda_{edge} := \lim_{\omega \rightarrow \Omega_{min}} \lambda(\omega) \in (0, \infty]. \quad (49)$$

⁴ When $\Omega_{max} < 2\Omega_{min}$ there are further gaps between the intervals $n[\Omega_{min}, \Omega_{max}]$. We do not consider frequencies lying in these gaps, since in all our examples the collective oscillation emerges below the band.

Third, the same construction applies to solutions which grow exponentially in time, $\delta\mathcal{F} \propto e^{\sigma t}$ with $\sigma > 0$, for which ω^2 is replaced with $-\sigma^2$ in Eq. (42). In this case $|R_n|$ decreases to zero when σ increases from zero to infinity, such that a growing solution exists if and only if $\lambda(0) > 1$. Hence, the linear stability of the steady state with respect to spherically symmetric perturbations requires that $\lambda(0) \leq 1$. For all the steady states of the reduced model which are analyzed in section VI we find $\lambda(0) < 0.6$.

Combining these properties, we arrive at the following criterion for a linearly stable steady state with $\lambda(0) < 1$. If $\lambda_{edge} > 1$, there exists an oscillating mode whose frequency ω_d lies below the frequency band and is determined by the equation

$$\lambda(\omega_d) = 1, \quad 0 < \omega_d < \Omega_{min}. \quad (50)$$

If $\lambda_{edge} < 1$, there are no oscillating modes with frequencies below the band. Further modes would appear if other eigenvalues of $S(\omega)$ exceeded one. This does not occur in our examples, in which the second largest eigenvalue remains smaller than 0.26 up to the edge of the band. Note that $1/\lambda_{edge}$ is the factor by which the perturbation of the potential in Eq. (36) would have to be multiplied for an oscillating mode to appear at the edge of the band; in this sense, λ_{edge} measures the strength of the self-gravity of the perturbation relative to the one that is required to prevent its mixing.

This criterion is a formulation in terms of action-angle variables of the Birman-Schwinger principle of Refs. [29, 30, 32, 34]. In fact, $\mu > 0$ is an eigenvalue of $\mathcal{K}(\omega)$ if and only if the linearized system in which $\delta\Phi$ is multiplied by the factor $1/\mu$ possesses an oscillating mode with frequency ω , and this is the property which characterizes the eigenvalues of the Birman-Schwinger operator of Ref. [34] with spectral parameter ω^2 . Therefore, $\lambda(\omega)$ coincides with the norm of the operator of Mathur defined in that reference, and the interval $0 < \omega < \Omega_{min}$ corresponds to the principal gap of the spectrum. In Refs. [29, 30, 34] the criterion is used to prove the existence or the absence of oscillating modes for certain classes of steady states. In section IV we describe a numerical method for the computation of $\lambda(\omega)$, which allows us to evaluate the criterion for steady states of arbitrary mass. [(Erik) As far as we could determine, this operator has not been evaluated numerically in the literature; Kunze's lecture notes [31] and Straub's thesis list it as an open task. Mathur's article [32] and Kunze's book [30] have to be checked before claiming this.]

D. Behavior at the edge of the frequency band

The value of λ_{edge} depends strongly on the way in which the DF of the steady state vanishes at the boundary of its support. To explain this, consider the reduced model and a steady state whose DF has a cutoff at the energy E_0 and behaves like

$$\mathcal{F}_{0,eq} \simeq \alpha(E_0 - E)^k, \quad E \rightarrow E_0, \quad (51)$$

with constants $\alpha > 0$ and $k > 1/2$, as is the case for the polytropes considered in Ref. [34]. Let J_{max} be the action variable of the orbit with energy E_0 , such that $\Omega_{min} = \Omega(J_{max})$ when Ω is a decreasing function of J . Then $\partial\mathcal{F}_{0,eq}/\partial J$ is proportional to $(J_{max} - J)^{k-1}$ near the edge, and Eq. (42) with $n = 1$ and $\omega = \Omega_{min}$ yields

$$|R_1(J; \Omega_{min})| \simeq \frac{1}{\Omega(J) - \Omega_{min}} \left| \frac{\partial\mathcal{F}_{0,eq}}{\partial J}(J) \right| \propto (J_{max} - J)^{k-2}, \quad J \rightarrow J_{max}. \quad (52)$$

This function is integrable at $J = J_{max}$ if and only if $k > 1$, which leads to the following two cases.

If $k > 1$, λ_{edge} is finite. Since $R(\omega)$ is proportional to the DF of the steady state, λ_{edge} is proportional to M_{gas}/M as long as this ratio is small enough for the contribution of Φ_{gas} to Φ_{eq} to be negligible. Therefore, for a gas of small mass one has $\lambda_{edge} < 1$ and there is no oscillating mode, in accordance with the damping result in Ref. [34]. When the mass of the gas increases, λ_{edge} grows, and an oscillating mode appears when it reaches the value one. The corresponding threshold mass is computed in section VI. Close to the edge one finds $\lambda_{edge} - \lambda(\omega) \propto (\Omega_{min} - \omega)^{\min\{k-1, 1\}}$, with a logarithmic correction when $k = 2$. Hence, slightly above the threshold the frequency of the mode satisfies $\Omega_{min} - \omega_d \propto (\lambda_{edge} - 1)^{1/(k-1)}$ when $1 < k < 2$, such that the closer k lies to one, the closer the frequency of the mode stays to the edge of the band.

If $1/2 < k \leq 1$, $\lambda(\omega)$ diverges for $\omega \rightarrow \Omega_{min}$, like $(\Omega_{min} - \omega)^{k-1}$ when $k < 1$ and like $|\log(\Omega_{min} - \omega)|$ when $k = 1$. Hence, $\lambda_{edge} = \infty$ and an oscillating mode exists for any mass of the gas, as proven in Ref. [34] for small masses. However, since the factor which multiplies the divergent term is proportional to M_{gas}/M , Eq. (50) implies that, to leading order in the limit $M_{gas}/M \rightarrow 0$,

$$\Omega_{min} - \omega_d \propto \left(\frac{M_{gas}}{M} \right)^{\frac{1}{1-k}} \quad (k < 1), \quad \Omega_{min} - \omega_d \propto \exp\left(-C \frac{M}{M_{gas}}\right) \quad (k = 1), \quad (53)$$

with a positive constant C . Therefore, when the mass of the gas is small the frequency of the mode lies extremely close to the edge of the band. Since a time of the order of $(\Omega_{min} - \omega_d)^{-1}$ is required to distinguish the mode from the response of the orbits close to the edge, which oscillates with a frequency close to Ω_{min} and decays algebraically (cf. Eq. (33)), in this regime the mode cannot be observed in a time evolution of practical duration.

When the angular momenta of the gas particles are distributed, the support of the steady state in the (J, L) -plane is two-dimensional, and in general the minimum Ω_{min} of the frequency is attained only at isolated points of its boundary. The additional integration over L in Eq. (43) then weakens the singularity of R_1 at the edge of the band, such that λ_{edge} can be finite also for $k \leq 1$, in which case the oscillating modes of the reduced model disappear when the mass of the gas is small enough. This is analyzed in section VII. **[(Erik) The precise statement depends on the family of steady states: for a cutoff in the energy and the isochrone or Kepler potential, Ω depends on L at fixed energy only through the self-gravity. To be settled together with section VII.]**

Finally, we comment on the case $\lambda_{edge} < 1$, in which the linearized system has no oscillating mode below the band. As we show in section VI, in this case the perturbation decays. When the self-gravity of the gas is not too weak, the response to an initial perturbation is dominated at intermediate times by a damped oscillation proportional to $e^{-i\omega t - \gamma t}$ with a frequency ω inside the band and a damping rate $\gamma > 0$, which is followed at late times by an algebraic tail of the form (33). The damped oscillation is the analogue of Landau damping in a plasma [27]. It does not correspond to an eigenvalue of the linearized system; instead, $\omega - i\gamma$ is a zero of the analytic continuation of the function $\omega \mapsto \det[1 - \mathcal{K}(\omega)]$ from the upper to the lower half of the complex plane, see for instance Refs. [1, 39, 40]. We do not compute this continuation in this article. Instead, we determine ω and γ by fitting the time series which are obtained from the numerical solution of the linearized equations and from the PIC code, as described in section IV.

IV. NUMERICAL METHODS

In this section we describe the numerical methods on which our study is based. We start with the PIC method used to solve the nonlinear Vlasov-Poisson system and with the construction of the self-consistent steady states. Next, we explain how the initial data are represented by the computational particles and how the response of the gas to a small perturbation is extracted from the simulations. Finally, we describe the numerical solution of the linearized equations, the computation of the gain $\lambda(\omega)$ of the self-consistent loop, and the determination of frequencies and damping rates from a time series.

A. Particle-in-cell method

In the PIC method [43, 44] the DF is represented by a finite number N of computational particles, each of which stands for a large number of gas particles. In spherical symmetry a computational particle is a spherical shell, which is characterized by its radius $r_j(t)$, its radial momentum $p_j(t)$, its angular momentum L_j and its mass m_j , $j = 1, 2, \dots, N$, the last two quantities being constant in time. This corresponds to the approximation

$$\mathcal{F}(t, r, p_r, L) \approx \sum_{j=1}^N \frac{m_j}{8\pi^2 L_j} \delta(r - r_j(t)) \delta(p_r - p_j(t)) \delta(L - L_j) \quad (54)$$

of the DF, where the factor $8\pi^2 L_j$ is due to the volume element (8). Introducing Eq. (54) into the Vlasov equation (5) yields the equations of motion

$$\frac{dr_j}{dt} = p_j, \quad \frac{dp_j}{dt} = \frac{L_j^2}{r_j^3} - \frac{\partial \Phi_{ext}}{\partial r}(r_j) - \frac{\partial \Phi_{gas}}{\partial r}(t, r_j) \quad (55)$$

for the computational particles; in the reduced model $L_j = L_0$ for all j . The first two terms on the right-hand side of the second equation are evaluated analytically, whereas the last one requires the solution of the Poisson equation (6). To this purpose we introduce the uniform grid $R_i := (i - 1/2)\Delta r$, $i = 1, 2, \dots, N_r$, which is staggered with respect to the center, and proceed in four steps.

- (i) The mass of each computational particle is assigned to its two neighboring grid points with the linear weights of the cloud-in-cell scheme [43, 44], such that the grid point R_i receives the mass

$$\mu_i := \sum_{j=1}^N m_j W\left(\frac{R_i - r_j}{\Delta r}\right), \quad W(y) := \max\{1 - |y|, 0\}. \quad (56)$$

- (ii) The density is approximated by the continuous, piecewise linear function $\rho(t, r)$ which assumes the values

$$\rho_i := \frac{\mu_i}{4\pi\Delta r(R_i^2 + \Delta r^2/6)} \quad (57)$$

at the grid points and which is constant for $0 \leq r \leq R_1$. The denominator in Eq. (57) is the integral of the function $4\pi r^2 W((r - R_i)/\Delta r)$ over r , which guarantees that the mass of the reconstructed density coincides with the total mass of the computational particles.

- (iii) The mass function $m(t, r)$ and the potential are obtained by integrating the equations $\partial m/\partial r = 4\pi r^2 \rho$ and $\partial \Phi_{gas}/\partial r = Gm/r^2$ outwards from the center. Since ρ is linear on each interval $[R_{i-1}, R_i]$, both integrals are computed in closed form, such that no further discretization error is introduced. The additive constant in Φ_{gas} is fixed by requiring that $\Phi_{gas} + r \partial \Phi_{gas}/\partial r = 0$ at the outermost grid point, which matches the potential to the exterior solution $-GM_{gas}/r$.
- (iv) The potential and its gradient are interpolated from the grid points to the positions of the computational particles with the same weights as in Eq. (56).

For particles which lie within the distance Δr from the center, the weights in steps (i) and (iv) are complemented by those of their mirror images at $-r_j$, in accordance with the regularity of the density at the center, and beyond the outermost grid point the exterior solution is used. Neither of these situations plays a role for the configurations considered in this article, in which the gas occupies a shell that lies well inside the grid.

Note that the potential which acts on a computational particle includes the contribution of the particle itself. In PIC schemes based on Cartesian coordinates the use of the same weights for the assignment and for the interpolation guarantees that a particle exerts no force on itself [43, 44]. In spherical symmetry this is not the case, and it should not be: a thin shell of mass m and radius r is subject to the mean value of the gravitational fields on its two sides, that is, to the force $-Gm/(2r^2)$ per unit mass. This is the self-gravity of the shell; the scheme reproduces it up to a correction of the order $\Delta r/r$, and we do not subtract it. In fact, we have verified that subtracting it reduces the order of convergence with respect to $1/N$ from two to one in the collapse of a cold homogeneous sphere, for which the exact solution is known.

Equations (55) are integrated in time with the fourth-order symplectic method of Yoshida [53], which is a composition of three steps of the second-order leapfrog method with step sizes $w_1 \Delta t$, $w_0 \Delta t$ and $w_1 \Delta t$, where $w_1 := 1/(2 - 2^{1/3})$ and $w_0 := 1 - 2w_1$. A leapfrog step with step size h consists of the update $p_j \mapsto p_j + hf_j/2$ of the momenta, with f_j the right-hand side of the second equation in (55), followed by the update $r_j \mapsto r_j + hp_j$ of the radii, the computation of the potential for the new radii, and a second update of the momenta with the new values of f_j . Hence, the Poisson equation is solved three times per time step. We use a symplectic method because, for the motion in a given potential, it does not introduce a secular drift of the energy and of the action variables of the particles, which would otherwise limit the accuracy of the observables at late times. For the same reason we keep the time step Δt fixed.

At selected times we compute the total energy (10),

$$\mathcal{E}_{gas} \approx \sum_{j=1}^N m_j \left[\frac{p_j^2}{2} + \frac{L_j^2}{2r_j^2} + \Phi_{ext}(r_j) + \frac{1}{2} \Phi_{gas}(t, r_j) \right], \quad (58)$$

the potential Φ_{gas} on the grid, from which $\delta\Phi$ is obtained according to Eq. (27), and the functions

$$h_n(t) \approx \sum_{j=1}^N m_j B(J_j, L_j) e^{-inQ_j}, \quad (59)$$

which follow from Eqs. (26,54). Here (Q_j, J_j) denote the action-angle variables of the particle j at the time t with respect to the reference potential, which are computed with the method described in Appendix A. In the reduced model we choose the weight function

$$B(J) = J^2 \exp \left[-\frac{(J - J_1)^2}{\sigma_1^2} \right], \quad (60)$$

with parameters J_1 and σ_1 which are specified below. The factor J^2 makes the function $B(J)e^{-inQ}$ continuous at the circular orbit for $|n| \leq 4$.

We have implemented this method in two codes, one for the reduced model and one for a general distribution of the angular momenta, which share the algorithm described above. Unless otherwise stated, the simulations of this article use $\Delta r = 0.1$, a grid which extends to $r = 20$, a time step $\Delta t = 0.05$ or 0.1 , and $N = 10^4$ computational particles.

[(Erik) To be decided: whether the codes and the scripts which reproduce the results are made public, in which case the repository should be cited here.]

B. Self-consistent steady states

As explained in section IID, the construction of a self-consistent steady state requires the solution of a nonlinear problem, because the DF is a prescribed function of integrals of motion which themselves depend on the potential generated by the gas. In the reduced model we consider the following two families of steady states. The first one consists of the polytropes

$$\mathcal{F}_{0,eq} = \alpha(E_0 - E)^k \quad \text{for } E < E_0, \quad (61)$$

and $\mathcal{F}_{0,eq} = 0$ for $E \geq E_0$, with an exponent $k > 1/2$. These are the steady states considered in Ref. [34]. The second family consists of the lowered Maxwellians

$$\mathcal{F}_{0,eq} = \alpha E_\gamma \left(g, \frac{E_0 - E}{\vartheta} \right) \quad \text{for } E < E_0, \quad E_\gamma(g, x) := \sum_{l=0}^{\infty} \frac{x^{g+l}}{\Gamma(g+l+1)}, \quad (62)$$

and $\mathcal{F}_{0,eq} = 0$ for $E \geq E_0$, see Ref. [54]. They include the King model [55] for $g = 1$, in which case $E_\gamma(1, x) = e^x - 1$, and the Wilson model [56] for $g = 2$, in which case $E_\gamma(2, x) = e^x - 1 - x$. Far from the cutoff the DF (62) is proportional to $e^{-E/\vartheta}$, whereas for $E \rightarrow E_0$ it vanishes like $(E_0 - E)^g$, such that g plays the role of the exponent k in Eq. (51). The parameter ϑ is specified through the dimensionless depth $W_0 := [E_0 - E_{min}(L_0)]/\vartheta$ of the potential well. In both families we prescribe the action variable J_{max} of the outermost orbit instead of the cutoff energy, which is then given by $E_0 = E(J_{max}, L_0)$, and we determine the constant α from the mass of the gas. Since the transformation to action-angle variables is symplectic, the latter is

$$M_{gas} = 16\pi^3 L_0 \int_0^{J_{max}} \mathcal{F}_{0,eq} dJ. \quad (63)$$

The steady state is computed by the following iteration, which starts with $\Phi_{gas} = 0$. Given Φ_{gas} on a radial grid, we first tabulate the function $J = A(E, L_0)/2\pi$ belonging to the potential $\Phi_{ext} + \Phi_{gas}$ by numerical quadrature (see Appendix A). This yields $E_{min}(L_0)$, the cutoff energy E_0 and, by inversion, the function $E(J, L_0)$. Next, we determine α from Eq. (63), compute the density from the second equation in (12), and obtain a new potential from Eq. (7). These steps are repeated until the maximum change of Φ_{gas} between two consecutive iterations is smaller than 10^{-13} . The radial grid used for this computation has the spacing 0.01, which is ten times smaller than the one of the PIC grid. The iteration converges geometrically, with a rate that deteriorates when the mass of the gas increases: for the lowered Maxwellians in the isochrone potential considered in section VI, the ratio between two consecutive changes of the potential grows from 0.02 for $M_{gas}/M = 0.0065$ to 0.59 for $M_{gas}/M = 0.5$, while for the polytropes in the Kepler potential between 6 and 42 iterations are required for $0.01 \leq M_{gas}/M \leq 1$.

For a DF with a distribution of angular momenta the same iteration is used, with the density computed from the second equation in (6), which requires the function $J(E, L)$ for each value of L in the support of the DF. **[(Erik) The families of steady states with a distribution in L are to be specified together with section VII.]**

C. Initial data and response to a perturbation

The initial data considered in this article consist of a self-consistent steady state and a small perturbation. In the reduced model they are of the form

$$\mathcal{F}_0(0, Q, J) = \mathcal{F}_{0,eq}(J) [1 + \varepsilon s(J) \cos Q], \quad s(J) := \left(\frac{J}{J_{max}} \right)^{3/2}, \quad (64)$$

where (Q, J) are the action-angle variables which belong to Φ_{eq} and $0 \leq \varepsilon \leq 1$ is the amplitude of the perturbation. The perturbation increases the density of the gas near the pericenters of the orbits and decreases it near their apocenters. It changes neither the mass of the gas nor the angle-average of its DF, and the factor $s(J)$ makes it regular at the circular orbit $J = 0$, where the angle variable is not defined.

If the radii and momenta of the computational particles were drawn at random from the DF (64), the fluctuations of the observables, which are of the order of $N^{-1/2}$, would exceed the signal produced by the perturbation for the numbers of particles used in this work. For this reason we proceed in a different way, which consists of three ingredients.

First, the computational particles are placed on a regular lattice in the action-angle variables of the steady state,

$$J_a := \left(a - \frac{1}{2}\right) \frac{J_{max}}{N_J}, \quad a = 1, \dots, N_J, \quad Q_b := \left(b - \frac{1}{2}\right) \frac{2\pi}{N_Q}, \quad b = 1, \dots, N_Q, \quad (65)$$

such that $N = N_J N_Q$, and their radii and momenta are obtained by inverting the map $(r, p_r) \mapsto (Q, J)$ numerically (see Appendix A). Since this map is symplectic, all the cells of the lattice have the same volume in phase space. Hence, the sums over the computational particles in Eqs. (58,59) are midpoint rules for the corresponding integrals over phase space, and they are free of the fluctuations of a random sample.

Second, the perturbation is encoded in the masses of the computational particles instead of their positions: the particle at the lattice point (Q_b, J_a) has the mass

$$m_{ab} = c \mathcal{F}_{0,eq}(J_a) [1 + \varepsilon s(J_a) \cos Q_b], \quad (66)$$

where the constant c is determined by the total mass M_{gas} . Since $\sum_b \cos Q_b = 0$, the perturbation does not alter the mass which is assigned to each orbit.

Third, each simulation with $\varepsilon > 0$ is accompanied by a reference simulation with $\varepsilon = 0$ which uses the same lattice and the same numerical parameters, and the response of the gas to the perturbation is defined by

$$\chi_n(t) := \frac{h_n^{(\varepsilon)}(t) - h_n^{(0)}(t)}{\varepsilon h_0^{(0)}(0)}, \quad \delta\Phi_{res}(t, r) := \frac{\delta\Phi^{(\varepsilon)}(t, r) - \delta\Phi^{(0)}(t, r)}{\varepsilon}, \quad (67)$$

where the superscripts refer to the amplitude of the perturbation. The subtraction of the reference simulation removes the errors which are common to both simulations, in particular the small departure of the discretized steady state from stationarity, which is quantified in section V. As long as the evolution is in the linear regime, the quantities (67) are independent of ε , and they can be compared directly with the solution of the linearized equations.

In plasma physics, a loading of the particles which avoids the fluctuations of a random sample is known as a quiet start [43], and particle methods in which the perturbation is carried by the weights of the particles have been used in Refs. [48, 49]. We stress that the lattice has to be constructed in the action-angle variables of the self-consistent potential Φ_{eq} . If those of the external potential are used instead, the resulting configuration is not stationary unless the mass of the gas is negligible. Furthermore, the regularity of the lattice is eventually lost due to the discreteness of the particles, which limits the time span over which the method is reliable; this time span is determined in section V.

For a distribution of angular momenta the lattice (65) is supplemented by the midpoints L_c of N_L equal subintervals of $[L_1, L_2]$, the action-angle variables being those of the corresponding effective potential, and the masses of the computational particles are proportional to $L_c \mathcal{F}(0, Q_b, J_a, L_c)$.

D. Numerical solution of the linearized equations

In order to compare the PIC simulations with the predictions of the linearized theory, we solve Eqs. (36,37) for the reduced model numerically as an initial-value problem. The function $\delta\mathcal{F}_0(t, Q, J)$ is represented by its values on a lattice of the form (65), whose points have the fixed radii $r_{ab} := r(Q_b, J_a)$, and it is advanced in time with the second-order splitting method of Strang [57]. Each time step consists of the update

$$\delta\mathcal{F}_0 \mapsto \delta\mathcal{F}_0 + \frac{\Delta t}{2} \frac{\partial \mathcal{F}_{0,eq}}{\partial J} \frac{\partial \delta\Phi}{\partial Q}, \quad (68)$$

followed by the exact solution of the transport equation $\partial_t \delta\mathcal{F}_0 + \Omega(J) \partial_Q \delta\mathcal{F}_0 = 0$ over the time Δt , which multiplies the n -th Fourier coefficient of $\delta\mathcal{F}_0$ with respect to Q by the factor $e^{-in\Omega(J)\Delta t}$, and by a second update of the form (68). The derivative with respect to Q is computed with the fast Fourier transform. The potential is obtained from the midpoint rule for Eq. (37),

$$\delta\Phi(t, r_{ab}) = -8\pi^2 G L_0 \Delta Q \Delta J \sum_{c=1}^{N_J} \sum_{d=1}^{N_Q} \frac{\delta\mathcal{F}_0(t, Q_d, J_c)}{\max\{r_{ab}, r_{cd}\}}, \quad (69)$$

with $\Delta Q := 2\pi/N_Q$ and $\Delta J := J_{max}/N_J$, which is the exact potential of a set of concentric shells. Since the radii r_{ab} do not change in time, the lattice points need to be sorted only once, and the evaluation of Eq. (69) at all of them requires a number of operations proportional to $N_J N_Q$. Unless otherwise stated we use $N_J = 1600$, $N_Q = 32$ and $\Delta t = 0.5$.

The initial datum is $\delta\mathcal{F}_0 = \mathcal{F}_{0,eq}(J)s(J)\cos Q$, which corresponds to Eq. (64) with $\varepsilon = 1$, and the output consists of $\delta\Phi$ and of the functions

$$\chi_n(t) = \frac{\sum_{a,b} \delta\mathcal{F}_0(t, Q_b, J_a) B(J_a) e^{-inQ_b}}{\sum_{a,b} \mathcal{F}_{0,eq}(J_a) B(J_a)}, \quad (70)$$

which are the counterparts of the quantities defined in Eq. (67). This method is free of particle noise, but by construction it is restricted to the linear regime.

E. Computation of the gain of the self-consistent loop

For the reduced model, the operator $S(\omega)$ defined in Eq. (48) is discretized as follows. The harmonics are truncated at $n \leq n_{max}$. The integral over J' in Eq. (43) is approximated by a Gauss-Legendre rule in the variable $\sigma \in (0, 1)$ defined by $J = J_{max}(1 - \sigma^2)$, with nodes J_a and weights w_a , $a = 1, \dots, N_J$. This substitution concentrates the nodes at the edge $J = J_{max}$ of the support, where R_n varies rapidly when ω is close to Ω_{min} . The integrals over the angles in Eq. (44) are approximated by the midpoint rule with N_Q points. In order to avoid the evaluation of the kernel for all pairs of points, we assign the points of each orbit to an auxiliary radial grid $\varrho_1 < \varrho_2 < \dots < \varrho_{N_g}$ with the linear weights W_i associated with its points, which gives

$$M_{nn'}(J_a, J_{a'}) \approx -4\pi GL_0 \sum_{i,i'=1}^{N_g} \frac{P_{(n,a)i} P_{(n',a')i'}}{\max\{\varrho_i, \varrho_{i'}\}}, \quad P_{(n,a)i} := \frac{2\pi}{N_Q} \sum_{b=1}^{N_Q} \cos(nQ_b) W_i(r(Q_b, J_a)). \quad (71)$$

The gain $\lambda(\omega)$ is then computed as the largest eigenvalue of the symmetric matrix with the entries

$$S_{(n,a)(n',a')}(\omega) = -\sqrt{w_a |R_n(J_a; \omega)|} M_{nn'}(J_a, J_{a'}) \sqrt{w_{a'} |R_{n'}(J_{a'}; \omega)|}, \quad (72)$$

which has the dimension $n_{max}N_J$, and the frequency ω_d of an oscillating mode is obtained by solving Eq. (50) with a bracketing root finder. Unless otherwise stated we use $n_{max} = 6$, $N_J = 200$, $N_Q = 128$ and $N_g = 2000$.

For steady states with $k > 1$ the result converges rapidly. Doubling each of these resolutions changes λ_{edge} by at most 2×10^{-5} for the Wilson model with $M_{gas}/M = 0.097$ in the isochrone potential, and by at most 2×10^{-4} and 3×10^{-3} for the polytropes with $k = 2$ and $k = 1.25$ in the Kepler potential, respectively. For $k \leq 1$, where λ_{edge} is infinite, the largest eigenvalue of the matrix (72) at $\omega = \Omega_{min}$ is finite and grows with the resolution; nevertheless, the frequency ω_d of the mode converges.

For a distribution of angular momenta we use quadrature nodes in the variables (E, L) , and the number of orbits is much larger. In this case it is more efficient to formulate the problem on the radial grid: writing the matrix with the entries $1/\max\{\varrho_i, \varrho_{i'}\}$ as CC^T by means of a Cholesky decomposition, the nonvanishing eigenvalues of the matrix (72) coincide with those of the matrix $4\pi G C^T P^T D(\omega) PC$, where $D(\omega)$ is the diagonal matrix which contains the products of $|R_n|$ with the quadrature weights. The dimension of this matrix is N_g , independently of the number of orbits.

F. Frequencies and damping rates

The frequency ω and the damping rate γ of the response are determined from the time series of $\chi_1(t)$, sampled at equidistant times, in the same way for the PIC simulations and for the solutions of the linearized equations. Within a given time window we fit the samples by a sum of K complex exponentials,

$$\chi_1(t) \approx \sum_{l=1}^K c_l e^{-i\omega_l t - \gamma_l t}, \quad (73)$$

using the matrix pencil method [58], and we identify (ω, γ) with the pair (ω_l, γ_l) of the term with the largest amplitude $|c_l|$. We use $K = 3$, and we estimate the uncertainty of the fit from the largest deviation among the results which are obtained with $K = 2$ and $K = 3$ for the given window and for the windows which are shortened by ten percent at either end. This quantity measures the uncertainty of the fit and not the numerical error of the underlying time series, which is assessed through the convergence tests of section V. For oscillating modes, we alternatively determine the frequency from the maximum of the periodogram of $\chi_1(t)$.

V. CODE VALIDATION

[(Erik) To be written.]

VI. RESULTS FOR FIXED ANGULAR MOMENTUM

[(Erik) To be written.]

VII. RESULTS FOR A DISTRIBUTION OF ANGULAR MOMENTA

[(Erik) To be written.]

VIII. CONCLUSIONS

[(Erik) To be written.]

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Appendix A: Numerical computation of the action-angle variables

In this appendix we describe the numerical evaluation of the action-angle variables defined in Eqs. (17,18,19) for a potential of the form $\Phi = \Phi_{ext} + \Phi_{gas}$, in which Φ_{gas} is only known on a radial grid. This is required for the construction of the self-consistent steady states, for the generation of the initial data and for the computation of the observables $h_n(t)$.

The values of Φ_{gas} on the grid are interpolated with a natural cubic spline, and beyond the last grid point we set $\Phi_{gas} = -GM_{gas}/r$. Given a point (r, p_r, L) we compute its energy E from Eq. (15), locate the minimum $r_{min}(L)$ of the effective potential V_L , and determine the turning points $r_1 < r_{min}(L) < r_2$ by solving the equation $V_L(r) = E$ with the bisection method on either side of the minimum. The integrand in Eq. (17) behaves like a square root at the turning points and the one in Eq. (18) like an inverse square root, which spoils the convergence of standard quadrature rules. We avoid this problem by means of the substitution

$$r = r_m + r_a \sin \theta, \quad r_m := \frac{r_1 + r_2}{2}, \quad r_a := \frac{r_2 - r_1}{2}, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}. \quad (\text{A1})$$

Since for a non-circular orbit the zeros of the function $E - V_L$ at the turning points are simple, one has $E - V_L(r) = r_a^2 \cos^2 \theta u(\theta)$ with a smooth and positive function u , and it follows that

$$A(E, L) = 2r_a^2 \int_{-\pi/2}^{\pi/2} \cos^2 \theta \sqrt{2u(\theta)} d\theta, \quad T(E, L) = 2 \int_{-\pi/2}^{\pi/2} \frac{d\theta}{\sqrt{2u(\theta)}}. \quad (\text{A2})$$

Both integrands are smooth, and we evaluate the integrals with a Gauss-Legendre rule with 64 nodes. The angle variable is obtained in the same way from

$$Q = \frac{2\pi}{T(E, L)} \int_{-\pi/2}^{\theta(r)} \frac{d\theta'}{\sqrt{2u(\theta')}}, \quad \theta(r) := \arcsin \left(\frac{r - r_m}{r_a} \right), \quad (\text{A3})$$

when $p_r \geq 0$, and from 2π minus the right-hand side of Eq. (A3) when $p_r < 0$. For the isochrone potential without self-gravity, for which the exact expressions (22,23) are available, this procedure yields the action variable with an error of the order of 10^{-14} and the period with an error of the order of 10^{-11} . Without the substitution (A1) the

corresponding errors are 2×10^{-6} and 9×10^{-3} , and they only decrease like the third and the first inverse power of the number of nodes, respectively.

For the inverse map $(Q, J) \mapsto (r, p_r)$ at fixed L we tabulate the function $J = A(E, L)/2\pi$ for 4000 values of the energy between $E_{\min}(L)$ and a value slightly above the cutoff energy, and obtain $E(J, L)$ by interpolation. Next, we determine the turning points and find, by bisection in θ , the point of the orbit at which the time elapsed since the passage through the pericenter is equal to $QT(E, L)/2\pi$ when $Q \leq \pi$ and to $(2\pi - Q)T(E, L)/2\pi$ otherwise. This gives the radius r , and the momentum is $p_r = \pm\sqrt{2(E - V_L(r))}$, with the positive sign when $Q \leq \pi$. Applying the direct map to the result reproduces the points of the lattice (65) with errors which are smaller than 10^{-8} in J and 5×10^{-7} in Q . The frequency is obtained from $\Omega = 2\pi/T(E, L)$.

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